Oracle SQL Assignment

Task 1: Doctor Table | Task 2: Employees Table

Task 1 — Doctor Table

1. Create table Doctor

```
CREATE TABLE Doctor (
    id NUMBER,
    name VARCHAR2(100),
    salary NUMBER,
    address VARCHAR2(200)
);
```

```
Table DOCTOR created.
```

2. Insert 10 records

```
INSERT ALL
    INTO Doctor VALUES (1, 'Ahmed', 5000, 'Cairo')
    INTO Doctor VALUES (2, 'Omar', 1000, 'Giza')
    INTO Doctor VALUES (3, 'Ali', 3000, 'Alexandria')
    INTO Doctor VALUES (4, 'Mona', 2000, 'Cairo')
    INTO Doctor VALUES (5, 'Sara', 7000, 'Giza')
    INTO Doctor VALUES (6, 'Youssef', 4000, 'Mansoura')
    INTO Doctor VALUES (7, 'Nada', 1000, 'Cairo')
    INTO Doctor VALUES (8, 'Khaled', 6000, 'Alexandria')
    INTO Doctor VALUES (9, 'Salma', 2000, 'Giza')
    INTO Doctor VALUES (10, 'Mahmoud', 3000, 'Cairo')
SELECT 1 FROM DUAL;
```

```
10 rows created.
```

3. Update record 3 salary to 20000

```
UPDATE Doctor
SET salary = 20000
WHERE id = 3;
```

```
1 row updated.
```

4. Delete record 9

```
DELETE FROM Doctor
WHERE id = 9;
```

```
1 row deleted.
```

5. Concatenate names with salary

```
SELECT name || ' - ' || salary AS name_salary
FROM Doctor;
```

```
NAME_SALARY
--------------------
Ahmed - 5000
Omar - 1000
Ali - 20000
Mona - 2000
Sara - 7000
Youssef - 4000
Nada - 1000
Khaled - 6000
Mahmoud - 3000

9 rows selected.
```

6. Display records with salary * 2

```
SELECT id, name, salary, salary * 2 AS double_salary, address
FROM Doctor;
```

Displays 9 records with an additional DOUBLE_SALARY column.

7. Salary is 1000, 2000, or 3000

```
SELECT *
FROM Doctor
WHERE salary IN (1000, 2000, 3000);
```

```
ID NAME SALARY ADDRESS
2 Omar 1000 Giza
4 Mona 2000 Cairo
7 Nada 1000 Cairo
10 Mahmoud 3000 Cairo

4 rows selected.
```

8. Rename Doctor to PRD_DOCTOR

```
RENAME Doctor TO PRD_DOCTOR;
```

```
Table DOCTOR renamed.
```

Task 2 — Employees Table

1. Create Employees table

```
CREATE TABLE Employees (
    EmployeeID NUMBER,
    FirstName VARCHAR2(50),
    LastName VARCHAR2(50),
    Department VARCHAR2(50),
    Salary NUMBER
);
```

```
Table EMPLOYEES created.
```

2. Insert the five records

```
INSERT ALL
    INTO Employees VALUES (101, 'John1', 'Doe1', 'HR', 20000)
    INTO Employees VALUES (102, 'John2', 'Doe2', 'IT', 50000)
    INTO Employees VALUES (103, 'John3', 'Doe3', 'CS', 40000)
    INTO Employees VALUES (104, 'John4', 'Doe4', 'IT', 10000)
    INTO Employees VALUES (105, 'John5', 'Doe5', 'ZX', 30000)
SELECT 1 FROM DUAL;
```

```
5 rows created.
```

3. Update EmployeeID 101 salary to 600000

```
UPDATE Employees
SET Salary = 600000
WHERE EmployeeID = 101;
```

```
1 row updated.
```

4. Delete where Department = 101

```
DELETE FROM Employees
WHERE Department = '101';
```

```
0 rows deleted — no inserted record has Department = '101'.
```

Alternative if the intended condition is EmployeeID = 101

```
DELETE FROM Employees
WHERE EmployeeID = 101;
```

```
1 row deleted — use this only if employee 101 should be deleted.
```

5. Retrieve employees in IT

```
SELECT *
FROM Employees
WHERE Department = 'IT';
```

```
EMPLOYEEID FIRSTNAME LASTNAME DEPARTMENT SALARY
102 John2 Doe2 IT 50000
104 John4 Doe4 IT 10000

2 rows selected if EmployeeID 101 has not been deleted.
```

6. Concatenate FirstName and LastName

```
SELECT EmployeeID,
       FirstName || ' ' || LastName AS FullName,
       Department,
       Salary
FROM Employees;
```

```
EMPLOYEEID FULLNAME DEPARTMENT SALARY
101 John1 Doe1 HR 600000
102 John2 Doe2 IT 50000
103 John3 Doe3 CS 40000
104 John4 Doe4 IT 10000
105 John5 Doe5 ZX 30000

5 rows before deleting EmployeeID 101.
```

Complete SQL Script

```
-- TASK 1
CREATE TABLE Doctor (
    id NUMBER,
    name VARCHAR2(100),
    salary NUMBER,
    address VARCHAR2(200)
);

INSERT ALL
    INTO Doctor VALUES (1, 'Ahmed', 5000, 'Cairo')
    INTO Doctor VALUES (2, 'Omar', 1000, 'Giza')
    INTO Doctor VALUES (3, 'Ali', 3000, 'Alexandria')
    INTO Doctor VALUES (4, 'Mona', 2000, 'Cairo')
    INTO Doctor VALUES (5, 'Sara', 7000, 'Giza')
    INTO Doctor VALUES (6, 'Youssef', 4000, 'Mansoura')
    INTO Doctor VALUES (7, 'Nada', 1000, 'Cairo')
    INTO Doctor VALUES (8, 'Khaled', 6000, 'Alexandria')
    INTO Doctor VALUES (9, 'Salma', 2000, 'Giza')
    INTO Doctor VALUES (10, 'Mahmoud', 3000, 'Cairo')
SELECT 1 FROM DUAL;

UPDATE Doctor SET salary = 20000 WHERE id = 3;
DELETE FROM Doctor WHERE id = 9;
SELECT name || ' - ' || salary AS name_salary FROM Doctor;
SELECT id, name, salary, salary * 2 AS double_salary, address FROM Doctor;
SELECT * FROM Doctor WHERE salary IN (1000, 2000, 3000);
RENAME Doctor TO PRD_DOCTOR;

-- TASK 2
CREATE TABLE Employees (
    EmployeeID NUMBER,
    FirstName VARCHAR2(50),
    LastName VARCHAR2(50),
    Department VARCHAR2(50),
    Salary NUMBER
);

INSERT ALL
    INTO Employees VALUES (101, 'John1', 'Doe1', 'HR', 20000)
    INTO Employees VALUES (102, 'John2', 'Doe2', 'IT', 50000)
    INTO Employees VALUES (103, 'John3', 'Doe3', 'CS', 40000)
    INTO Employees VALUES (104, 'John4', 'Doe4', 'IT', 10000)
    INTO Employees VALUES (105, 'John5', 'Doe5', 'ZX', 30000)
SELECT 1 FROM DUAL;

UPDATE Employees SET Salary = 600000 WHERE EmployeeID = 101;

-- Literal interpretation: Department = 101
DELETE FROM Employees WHERE Department = '101';

SELECT * FROM Employees WHERE Department = 'IT';
SELECT EmployeeID, FirstName || ' ' || LastName AS FullName,
       Department, Salary
FROM Employees;
```